

Observed PM_{2.5} at an Identified U.S. Department of State Monitor in Dhaka, 2019-February 2025

A provenance audit, robust trend assessment, and short-horizon forecast

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Awnon Bhowmik¹, Mahmudul Hasan², and Goutam Saha^{2,3*}

¹ College of Engineering & Computer Science, Colorado Technical University, Colorado Springs, CO 80907, USA ² Department of Mathematics, University of Dhaka, Dhaka 1000, Bangladesh ³ Miyan Research Institute, International University of Business Agriculture and Technology, Uttara, Dhaka 1230, Bangladesh * *Correspondence:* gsahamath@du.ac.bd

Abstract

Air-quality studies can be invalidated when modeled concentrations, scraped indexes, changing stations, and repeated monthly values are treated as one observational series. We audited the data lineage of a Dhaka air-quality repository and rebuilt the analysis from official U.S. EPA AirNow archive records for the U.S. Department of State monitor DK1010001 (23.796374 N, 90.424614 E). The archive supplied 2,209 valid preliminary 24-hour PM_{2.5} summaries from 2019-01-01 to 2025-03-24. Using a $\geq 75\%$ valid-day rule and excluding the truncated terminal month yielded 71 complete months from January 2019 through February 2025. The mean of complete monthly means was 98.65 $\mu\text{g}/\text{m}^3$ (median 79.66). Seasonal Mann-Kendall indicated an increase ($p=0.0046$), and month-adjusted HAC regression estimated 3.14 $\mu\text{g}/\text{m}^3/\text{year}$ (95% CI 1.42 to 4.86); however, trend-free prewhitening was not significant ($p=0.18$), so trend inference is sensitive to serial dependence. A season-matched March-August 2020 contrast was -7.32 $\mu\text{g}/\text{m}^3$ (95% bootstrap CI -42.90 to 28.65); the design does not support a causal lockdown claim. SARIMA achieved the lowest rolling-origin MASE (0.85) and was used for a 24-month forecast; 2030 values are shown only as conditional targets/benchmarks. PM₁₀, NO₂, and SO₂ remain important AQI pollutants in principle, but their legacy columns lacked the station, unit, and method

continuity required for empirical analysis. The audit reverses the legacy declining trend claim and removes unsupported multi-pollutant, source-apportionment, health-burden, and citywide-exposure conclusions.

Keywords: PM2.5; Dhaka; AirNow; data provenance; seasonal Mann-Kendall; rolling-origin forecasting; reproducibility

1. Introduction

Air pollution is a major environmental-health concern, and WHO guidance covers PM2.5, PM10, NO2, SO2, ozone, and carbon monoxide because their health effects and regulatory averaging periods differ [1]. Prior Dhaka studies using other monitoring periods or research designs have reported strong winter seasonality, high particulate concentrations, and roles for gaseous pollutants [6-10]. Those studies establish the scientific relevance of a multi-pollutant perspective; they do not validate unrelated columns in the present repository.

A credible time-series study requires consistent measurement lineage. The original manuscript described a 2017-2025 multi-pollutant monitoring record, but audit of its code, workbooks, and Git history found CAMS reanalysis values, scraped AQI, monthly values repeated across days, unresolved station identities, and annual contextual variables repeated monthly. The original 41-reference list also contained wrong or unrelated DOIs; its retained sources were checked against publisher, registry, or official metadata.

This revision asks: (1) what seasonal and temporal patterns are supported by one identified ground monitor; (2) how sensitive is trend inference to seasonality and serial dependence; (3) what non-causal association is visible during the 2020 restriction period; and (4) which short-horizon forecast model outperforms seasonal naive in rolling tests? The scope is deliberately narrower than the environmental importance of Dhaka's full pollutant mixture.

2. Data and methods

2.1 Monitor and provenance

The primary source is the official dated AirNow daily_data_v2.dat archive. The original provider is U.S. Department of State Bangladesh - Dhaka; AirNow is the distribution system. Each source row reports station DK1010001 (full code 050DK1010001), site name Dhaka, latitude 23.796374, longitude 90.424614,

parameter PM2.5-24hr, unit UG/M3, duration 24 hours, concentration, AQI, and category. The archive row does not encode instrument metadata, so the instrument remains unresolved. AirNow values are preliminary and subject to revision.

The downloader used verified HTTPS and preserved every URL, exact source line, response SHA-256, and retrieval timestamp. Bangladesh Department of Environment reports were selected as a separate validation source, but no reproducible raw machine-readable DoE series was found; no numeric series was extracted from charts or spliced into AirNow.

2.2 Pollutant scope and AQI interpretation

PM10, NO2, and SO2 were not removed because they are unimportant. A multi-pollutant AQI calculates a subindex from each simultaneous physical concentration and reports the maximum. Any of these pollutants can therefore determine AQI when its subindex is largest. They are parallel inputs to the AQI algorithm, not causal covariates that “influence” an independently measured AQI.

Pollutant	Legacy evidence	Revised empirical treatment
PM2.5	Source-mixed CAMS and monthly values	Replaced by identified AirNow/DoS ground-monitor observations
PM10	Modeled/repeated values; station and method unresolved	Excluded pending a homogeneous physical concentration series
NO2	Modeled/repeated values; units and station unresolved	Excluded pending a homogeneous physical concentration series
SO2	Modeled/repeated values; units and station unresolved	Excluded pending a homogeneous physical concentration series

Because PM2.5 is the only verified simultaneous concentration, the recalculated AQI in this study is specifically the PM2.5 subindex. The stored `dominant_pollutant=pm25` value means PM2.5 is the only available calculated subindex; it does not prove that PM2.5 dominated Dhaka’s complete multi-pollutant AQI on every day. Prior Dhaka studies confirm that PM10 and gaseous

pollutants warrant monitoring [6,7], but their results cannot be silently spliced into this station series.

2.3 QA, aggregation, and cutoff

The pipeline rejected wrong station, agency, pollutant, unit, duration, duplicates, negative values, and dates beyond retrieval. Asia/Dhaka defines local days. Monthly summaries require $\geq 75\%$ valid daily AirNow summaries. Hourly completeness is unknown and remains null. A terminal month must reach month-end, so March 2025 (24/31 days) is partial and the cutoff is February 2025. No missing observation was imputed.

AQI was recalculated from physical PM_{2.5} using U.S. EPA breakpoints effective 6 May 2024. Source-reported AQI was retained separately because historical values used contemporaneous breakpoints. AQI is a mathematical transform here, not an independent environmental outcome.

2.4 Statistical analysis

Descriptive results include valid counts, mean, median, standard deviation, and IQR. Monthly climatology includes n and t -based 95% confidence intervals. Seasonal differences use Kruskal-Wallis, epsilon-squared, and Holm-corrected pairwise Mann-Whitney tests.

Trend methods comprise seasonal Mann-Kendall with tie correction, same-month Sen slope with a seeded year-block bootstrap interval, trend-free prewhitening, and elapsed-time regression with calendar-month fixed effects and 12-lag Newey-West covariance. Pettitt's test is a change diagnostic only.

COVID analyses compare March-August 2020 with the same months of 2019 and fit a month-adjusted interrupted association. Meteorology was unavailable; language is associational rather than causal. Annual means are compared with annual guidelines, and daily observations with the specified 24-hour forms.

2.5 Forecasting and scenarios

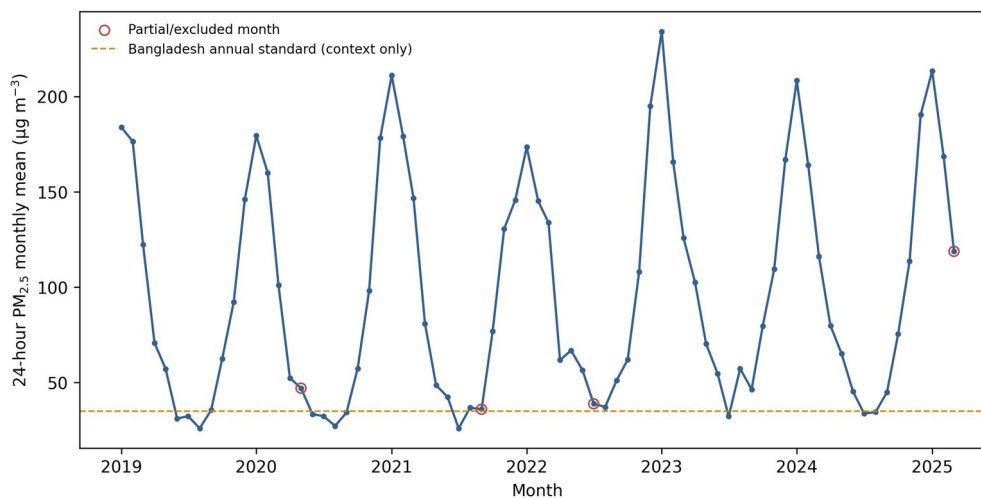
Seasonal-naive, naive, drift, ETS, SARIMA, and time/month regression models were evaluated at three expanding origins with 12-month test horizons. Metrics were MAE, RMSE, MASE, sMAPE, and 95% interval coverage. Positive-support statistical models used log_{1p} concentration. The lowest mean MASE selected the model. The empirical horizon is 24 months. 2030 lines are deterministic

benchmarks, not forecasts; no probabilistic interval is fabricated for policy implementation.

3. Results

3.1 Completeness and distribution

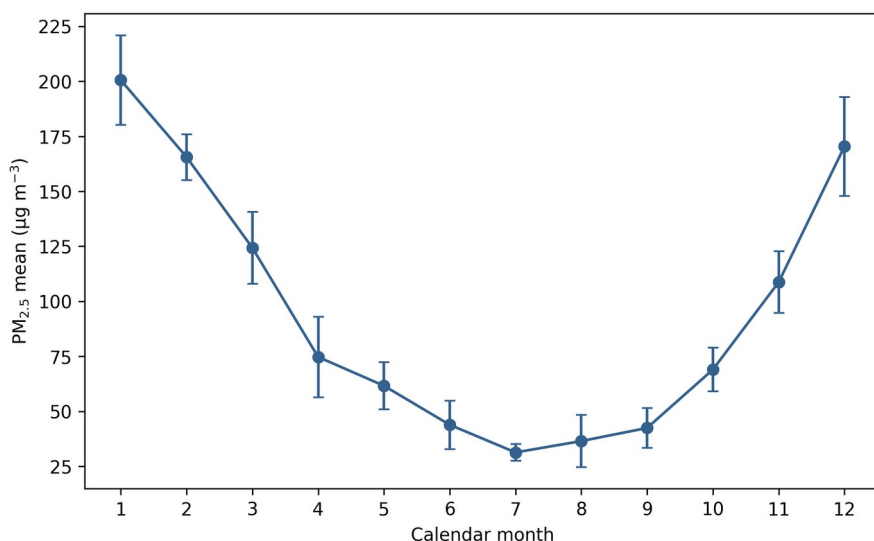
The observed archive contained 2,209 daily summaries. Of 75 calendar months, 71 met the primary rule. Complete-month PM_{2.5} ranged from 25.97 to 234.06 ug/m³; the IQR was 98.82 ug/m³.



Observed monthly PM_{2.5}. Open symbols are excluded partial months. Source: U.S. Department of State via AirNow.

3.2 Seasonality

Calendar-month climatology peaked in month 1 at 200.53 ug/m³ and was lowest in month 7 at 31.33 ug/m³. Seasonal differences were large (Kruskal-Wallis epsilon-squared 0.86, p<0.001).



Monthly climatology with 95% confidence intervals; complete months only.

3.3 Trends and break diagnostic

Seasonal Mann-Kendall yielded $z=2.83$, $p=0.0046$. The seasonal Sen estimate was 2.80 ug/m3/year (95% bootstrap CI -0.00 to 4.88). HAC regression estimated 3.14 (95% CI 1.42 to 4.86), but prewhitening yielded $p=0.18$ with lag-1 autocorrelation 0.80. We therefore interpret the direction as suggestive of an increase but not robust across dependence adjustments. Pettitt's test did not identify a significant single change ($p=0.48$).

3.4 2020 restriction-period association

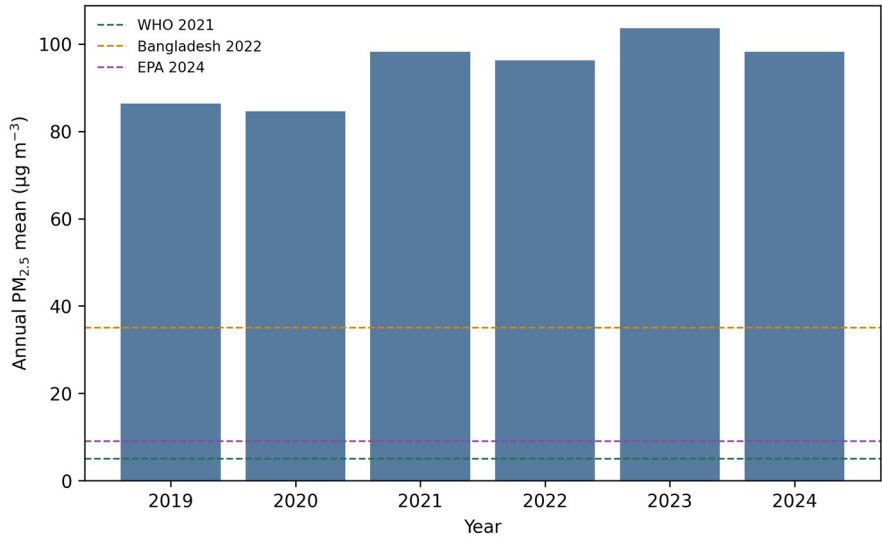
The season-matched difference was -7.32 ug/m3 (95% CI -42.90 to 28.65, $p=0.93$). The month-adjusted interrupted coefficient was -10.62 ug/m3 (95% CI -14.88 to -6.35). The discrepancy and lack of meteorological adjustment preclude a causal conclusion.

3.5 Guideline comparisons

All complete annual monitor means (2019-2024) exceeded WHO 2021 (5 ug/m3), Bangladesh 2022 (35 ug/m3), and U.S. EPA 2024 (9 ug/m3) annual levels. EPA comparisons are descriptive because formal NAAQS determinations use design-value procedures and regulatory data.

Year	PM2.5 mean (ug/m3)	valid-day coverage (%)
2019	86.4	98.4

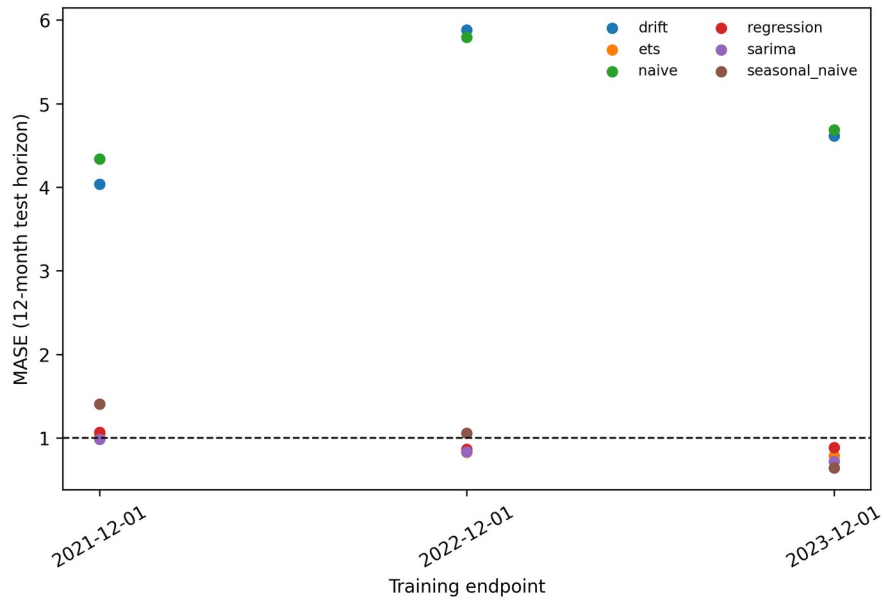
Year	PM2.5 mean (ug/m3)	valid-day coverage (%)
2020	84.6	96.2
2021	98.2	94.5
2022	96.3	94.8
2023	103.6	99.2
2024	98.2	98.9



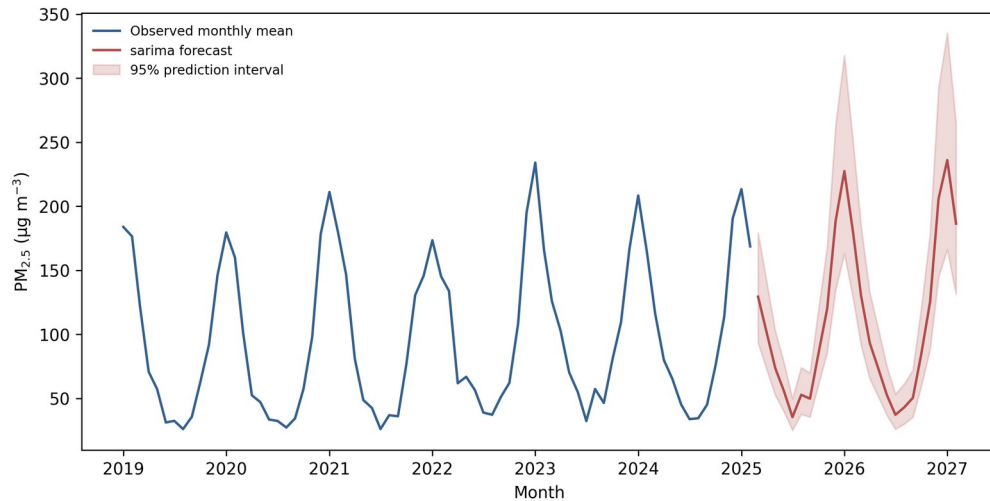
Annual monitor means with annual reference levels; these are not interchangeable with 24-hour standards.

3.6 Forecast validation

The selected model was sarima (mean MAE 13.64, RMSE 18.05, MASE 0.85, sMAPE 14.8%). Seasonal naive had MASE 1.03.

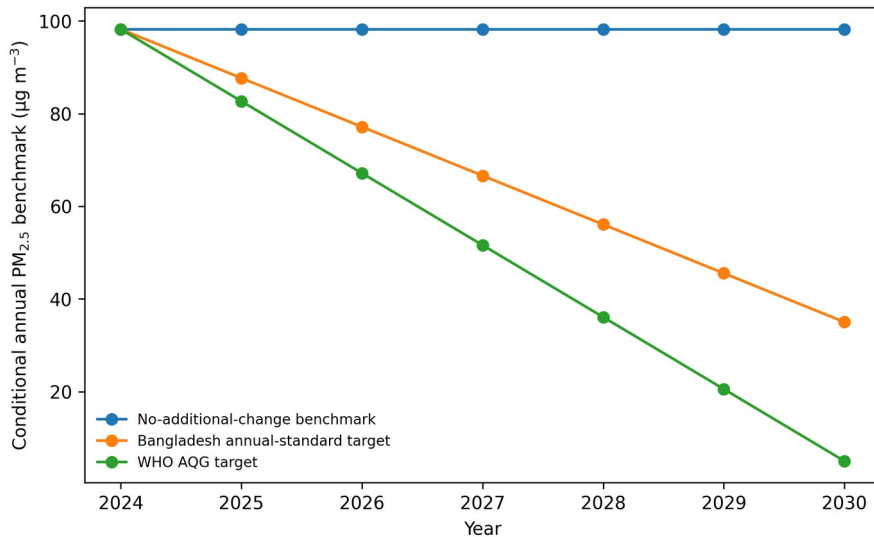


Rolling-origin 12-month backtesting.



Selected 24-month forecast with 95% prediction intervals.

The 2030 benchmarks end at 98.19 $\mu\text{g/m}^3$ for no additional change, 35 $\mu\text{g/m}^3$ for the Bangladesh annual standard, and 5 $\mu\text{g/m}^3$ for the WHO AQG. These values express assumptions/targets, not expected outcomes.



Conditional 2030 benchmark paths, not forecasts.

4. Discussion

4.1 What changed after the provenance audit

Rebuilding from one identified monitor changes both scope and inference. The legacy analysis reported a strong PM_{2.5} decline using a source-mixed 2017-2025 series. The audited 2019-2025 monitor record instead suggests an increase under seasonal MK and HAC regression, with loss of significance after prewhitening. The correct conclusion is sensitivity, not a definitive monotonic trajectory.

4.2 Why PM₁₀, NO₂, and SO₂ still matter

The original paper was correct that PM₁₀, NO₂, and SO₂ are environmentally and regulatorily important. Historical multi-site work in Dhaka measured PM₁₀ and gaseous pollutants alongside PM_{2.5} and found pollutant-specific seasonal and trend behavior [6,7]. Their absence from the revised numerical tables is an evidence boundary: the repository does not contain a homogeneous, traceable series for them. Treating modeled CAMS values, repeated monthly values, and unresolved ground records as one series would produce precise-looking but invalid pollutant rankings and forecasts.

This distinction also changes AQI interpretation. If simultaneous PM₁₀, NO₂, SO₂, CO, and O₃ concentrations become available, every applicable subindex should be calculated under one declared standard and the daily maximum should be

retained. Until then, the present calculated AQI is a PM2.5 subindex, and no claim is made about the historical percentage of days dominated by other pollutants.

4.3 Seasonality, sources, COVID, and forecasting

Seasonality is much more stable: winter concentrations are high and monsoon concentrations low, consistent with earlier Dhaka research [6-10]. Published black-carbon and isotope studies support roles for local and regional combustion sources [9,10], but the present data do not apportion sources. PM2.5 cannot be summed with PM10 for source apportionment, and no chemical speciation, PMF/CMB, inventory, or transport attribution is implemented here.

The season-matched COVID comparison is uncertain. Although the interrupted coefficient is negative, a single-station ecological time series without meteorology or mobility controls cannot isolate lockdown effects. Likewise, AQI-PM2.5 correlation would be definitional and is not presented as discovery. The selected 24-month forecast is an empirical extrapolation, whereas the 2030 lines are policy/health benchmarks; Bangladesh's national air-quality plan provides relevant policy context but does not make those benchmark paths probabilistic forecasts [11].

5. Limitations

One monitor does not measure citywide spatial exposure. Values are preliminary daily summaries; per-hour completeness and archive-row instrument metadata are unavailable. Three internal months with 66-71% coverage are excluded from trend inference but retained explicitly in forecasting to preserve the calendar without imputation. No independent numeric DoE validation or meteorological series was available. The absence of traceable PM10, NO2, SO2, CO, and O3 prevents calculation of a complete multi-pollutant AQI and pollutant-dominance fractions. The feed cessation limits the empirical period. Forecast validation has only three annual rolling origins, and the 24-month outlook cannot establish 2030 conditions. Trend conclusions depend on serial-correlation handling. The supplied original contained no tracked-change history; this is therefore a clean revision accompanied by claim and reference audits.

6. Conclusion

The defensible result is a narrow, station-specific PM2.5 study. Pollution at this monitor remained far above annual health and regulatory reference levels, with

strong seasonality and method-sensitive trend evidence. The data do not support multi-pollutant trends, citywide exposure, source apportionment, an EKC, precise attributable mortality, or a causal lockdown claim. A 24-month forecast can be reported with rolling-origin validation; 2030 values remain conditional benchmarks. PM10, NO2, and SO2 remain essential monitoring targets and potential AQI-determining pollutants; defensible analysis of them requires a new, station-identified, unit-verified, QA-documented concentration series.

Data and code availability

Exact extracted AirNow rows, request URLs, response checksums, standardized products, source manifest, provenance ledger, code, tests, and generated outputs are in the repository. Large or terms-restricted CAMS, BMD, OpenAQ, or future DoE raw files must be downloaded separately under provider terms. The paper was generated from repository commit a23c18f84355f8d87cc17e81d9542b7fc9879e7f. Raw retrieval occurred on 2026-07-18 UTC (2026-07-17 America/New_York). The supplied original DOCX is preserved unchanged with a SHA-256 manifest; its complete claim and 41-reference audits are distributed with the revision.

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